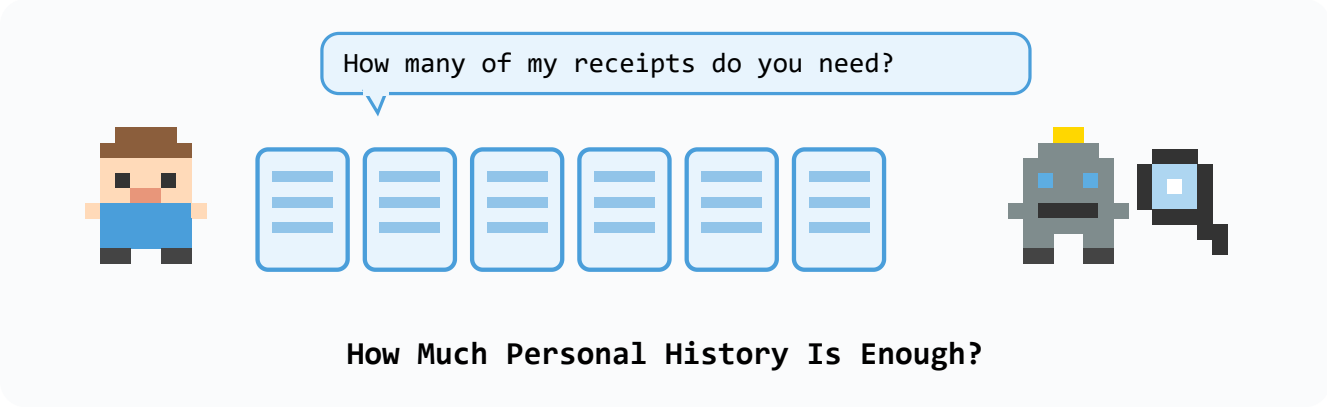


How Much Personal History Is Enough?

A research project explained so simply that anyone can follow it, from the first question to the last number.

The short answer. On real bank data, a computer "detective" that learns your spending habits needs **10 to 22** of your own transactions before it is nearly as good as it will ever get. (About 20 if it also studies the dates of your payments, about 10 if it doesn't.) Before that, a detective that learned from *other* people does better, so start with the crowd and switch to you once you have enough history. And a detective that is good at *spotting* odd transactions still needs at least **19** of your transactions before it can set a fair alarm bell. In other kinds of spending data the number is very different, so there is no single magic number.

1. The question, as a story

Maya opens a new bank account. Her bank wants a computer program that notices when something on her account looks *unusual for Maya*, for example a payment that doesn't look like something she would make. The program learns what is normal by looking at Maya's own past transactions (her "receipts").

But on day one Maya has **zero** receipts. After a week she has a few. After a year she has hundreds. So the question is simple:

How many of Maya's own receipts does the program need before its idea of "normal Maya" is good enough to use?

Real systems pick a number (for example "ignore users with too little history"), but nobody had carefully measured what that number should be. That is what this project measures.

Why it matters

- **New customers, young people and people who just moved country** have the shortest histories. If the program is bad for them, they get treated unfairly.
- A **false alarm** can mean a blocked card on holiday or a frozen account. So a program that "cries wolf" is not harmless.
- Keeping years of someone's transactions is a **privacy** cost. If a short history is enough, we can keep less.

> **Important:** "unusual" is **not** the same as "fraud". Most unusual transactions are innocent: a holiday, a new job, a house move. This project measures how well a program spots *unusual-for-you* behaviour. It does not catch criminals and it says nothing about anyone being guilty of anything.

2. What we expected (the hypotheses)

#	Our guess before the experiments	What we found
H1	The program gets better quickly at first, then stops improving (a "plateau").	**Yes** on the bank data: big gains up to about 10-22 receipts (depending on whether the detective looks at dates), only a little after 50.
H2	With very few receipts, a program trained on *other people* beats one trained on you.	**Yes**: below about 10 receipts the crowd wins.
H3	A blend that trusts you more as your history grows gets the best of both.	**Only at the start.** With very few receipts the blend beats the "you only" detective. After that, the "you only" detective was as good or better in most of our tests, so it is safer to switch to it. In the online shop, where the crowd knew little, the blend was worse even at the start.
H4	Being good at ranking odd transactions also means having a sensible alarm.	**No**: the alarm needs its own fair rule, and at least 19 receipts.
H5	People with messier habits need more receipts.	**Between very different kinds of people, yes.** In the sandbox, messy *types* of people need about ten times more receipts than tidy types. **In the real bank data, no:** how messy a person's receipts look did not tell us who needs more.
Q	(An open question rather than a guess) Is the answer the same number in every kind of data?	**No**: the online shop and the card factory give very different answers. Each system has to measure its own.

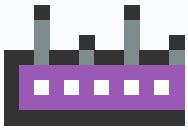
3. The data: where the receipts come from



Czech bank
real, 1993-98



Online shop
real, 2009-11



Card factory
computer-made

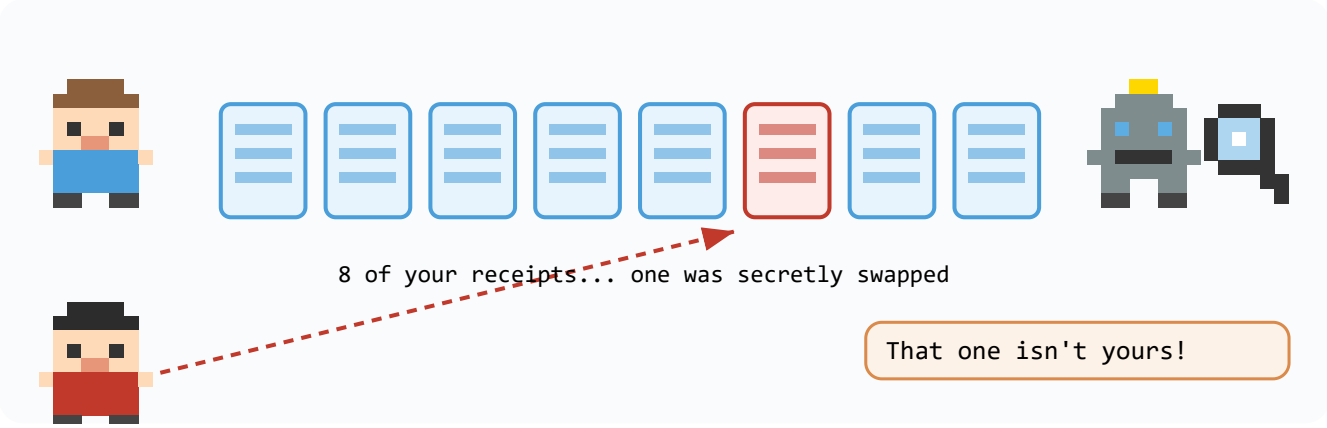


Sandbox
we set the rules

World	Real or made-up?	What is in it	Why we used it
Czech bank (the "Berka" data)	Real, 1993-1998	About 1 million transactions from 4,500 bank accounts: date, amount, credit or debit, type of payment.	The main test: real people, long histories.
Brand-new accounts (same bank)	Real	The *first* 110 transactions of 3,727 accounts.	The real "new customer" situation the question is about.
Online shop (Online Retail II)	Real, 2009-2011	Every item bought from a UK online gift shop: who, when, which product, how many, what price.	A completely different kind of spending, to see whether the bank results hold elsewhere.
Card factory (TabFormer)	Made by a computer program	24 million fake card payments with shop names, shop types and places.	Has details the bank data lacks (which shop, where) and very long histories.
Sandbox (our simulation)	Made by us	240 pretend people whose habits we set on purpose: tidy, messy, travellers, people whose habits change.	The only way to change "how messy someone is" and watch what happens.

> We never mix the worlds together. A made-up dataset can show that something *can* happen; only real data can show that it *does*.

4. The impostor game: how we make "unusual" transactions



None of these datasets says which transactions are unusual. So we create unusual transactions in a way that cannot cheat:

1. Take a person's 60 most recent receipts. This is the **test pile**.
2. Secretly swap 6 of them (10%) for **real receipts from a different person** (an "impostor"). The swapped receipt keeps the original date, so the timing does not give it away.
3. Ask the detective to rank all 60 from "most unusual" to "most normal". A good detective puts the 6 impostor receipts near the top.

Because we *know* which 6 we swapped, we can score the detective fairly. Nobody's opinion decides what is "unusual": the label is simply "this receipt came from someone else".

We also play two variations: **swap only receipts the customer made themselves** (bank fees and interest are left alone), and **make one amount five times bigger** instead of swapping.

5. A fair race: the fixed test pile

To see how much history helps, we give the same detective more and more of a person's past receipts: 1, 5, 10, 12, 15, 18, 20, 22, 25, 50, 100, 250. The trick that keeps this fair: **the test pile never changes**. Each detective, whatever it learned from, is tested on exactly the same 60 receipts. So any change in score comes from the amount of history, not from an easier or harder test.

older receipts (unused)	the *n* receipts the detective learns from	the 60-receipt test pile (always the same)

The detective never sees a test receipt while learning, and it never sees which receipts were swapped.

6. The detectives

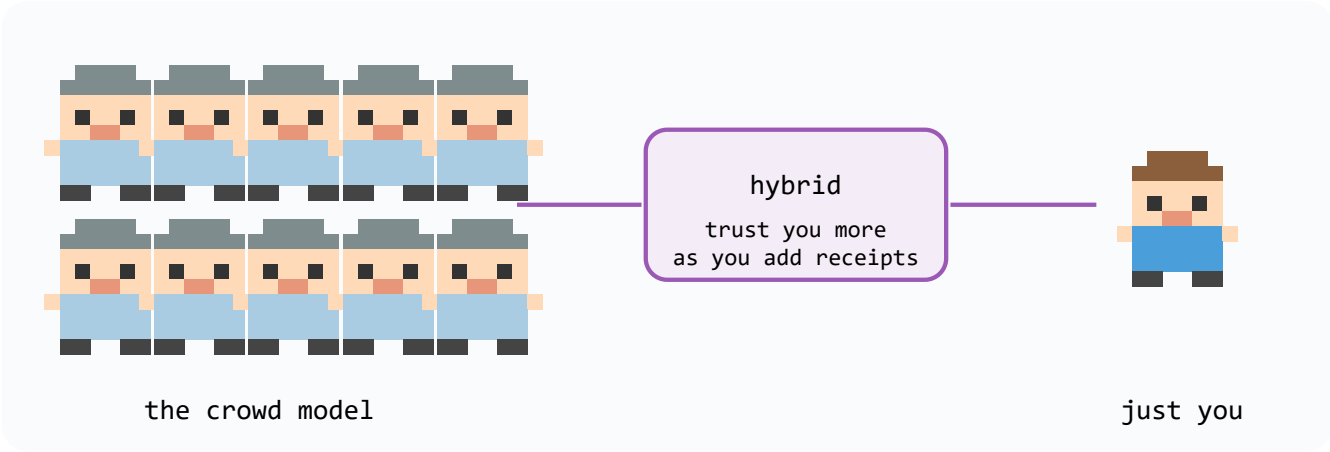
We tried eight different "detectives" (anomaly detectors). Each has a different idea of "unusual":

Detective	Its idea of "unusual", in plain words
Gaussian *z* and robust *z*	"Is the **amount** much bigger or smaller than usual?" Looks at one thing only.
Mahalanobis distance	"How far is this receipt from your **typical** receipt, counting all the details at once and how they usually go together?" (Like noticing that a big payment on a Tuesday is odd for you, even if big payments and Tuesdays are each normal.)
Mahalanobis, Ledoit-Wolf version	The same idea, with a more cautious way of estimating "how details go together" when there is little data.
k-nearest neighbours	"Is this receipt far from your 5 most similar past receipts?"
Local outlier factor	"Is this receipt in a lonelier spot than your other receipts?"
Isolation forest	"How quickly can I separate this receipt from the rest with random yes/no questions?" Odd ones are separated quickly.
Autoencoder	A tiny neural network that learns to "redraw" your receipts. If it redraws a receipt badly, the receipt is unusual.

How we score a detective

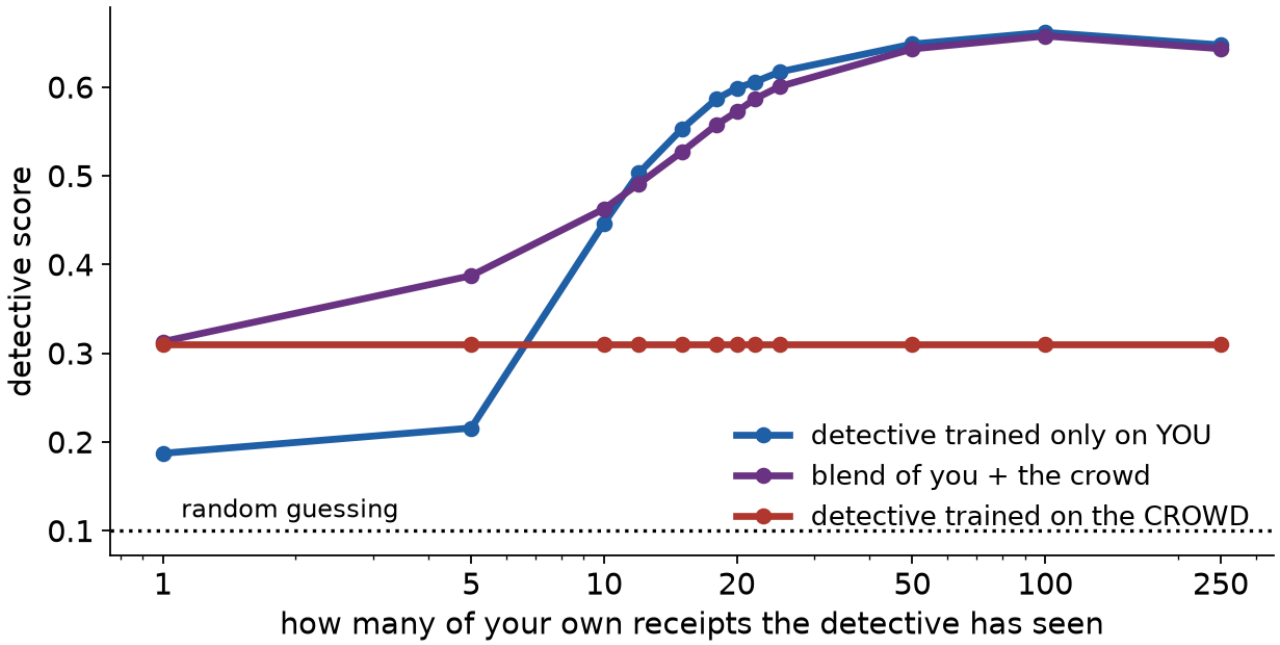
We use a score called **PR-AUC**. Think of it as: "if you read the detective's list from the top, how many of the swapped receipts do you find early, and how few honest ones do you trip over?" Random guessing scores about **0.10** (because 10% of the pile is swapped). A perfect detective scores **1.00**.

7. You, the crowd, and the blend



- **Personal detective:** learns only from your receipts.
- **Crowd detective:** learns from 400 *other* people (never from you, and never from anyone being tested).
- **Blend (the "hybrid"):** starts as the crowd detective and slides toward the personal one as you add receipts. The slider is simple: with n receipts, your own data gets weight $n/(n+25)$. With 25 receipts it is a 50/50 mix. We chose 25 before seeing any results so we could not tune it to look good, and then checked 5, 10, 50 and 100 too.

8. Result 1: a little history goes a long way, then stops helping



Real bank data, 300 accounts, averaged over 5 different groups of accounts and 20 rolls of the impostor game each.

With **1** receipt the personal detective scores 0.19 (barely better than guessing). With **10** it scores 0.45, with **20** it scores 0.60, and its best is about 0.66.

It reaches **90% of its best at 20 receipts** in 3 of the 5 groups of accounts and 22 in the other 2 (re-sampling one group's accounts 1,000 times gives 18 to 20 receipts 95% of the time). Going from 50 to 250 receipts adds almost nothing.

The date clue

Part of the detective's skill on the bank data came from **dates**. Some receipts are made by the bank itself (interest, fees), and the bank posts those on fixed days of the month. When such a receipt was swapped into someone else's pile on an ordinary day, a detective that studies dates could notice the odd date. Learning your date habits takes more receipts, which is why it needed about 20. When we only swapped receipts that customers made themselves, the date clue stopped helping, and a detective that ignores dates reached

90% of its best after only about 10 receipts, just as good in the end. We think this explanation is likely, but we did not test it directly.

Which detective was best?

On the bank data the simple Mahalanobis detective was the best of the full detectives from 10 receipts onward. The tiny neural network learned just as fast but topped out lower (0.54). The neighbour detective needed about 50 receipts before it was any use, and the fancier isolation forest was one of the weakest, never scoring above 0.35. But in the computer-made card factory the isolation forest was the **best** detective from 5 to 100 receipts, and in the online shop the simplest detective of all (just "is the amount unusual?") beat every fancy one. So "which detective is best" depends on the data. There is no universal winner.

9. Result 2: early on, the crowd knows you better than you do

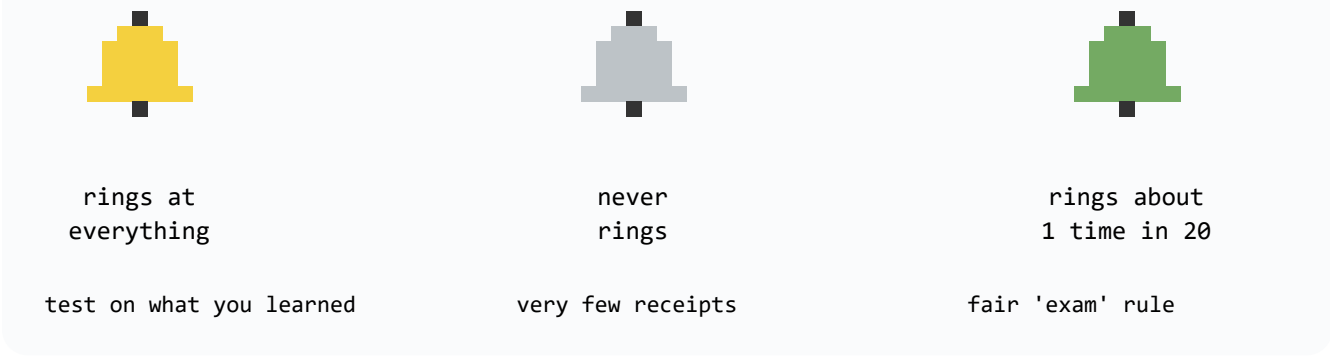
The crowd detective scores 0.31 no matter how many of your receipts exist, because it never looks at them. The personal detective only beats it from about **10** receipts (the same answer in all 5 groups: 10, 10, 10, 10, 10).

The blend starts at the crowd's level (0.31 at 1 receipt) and joins the personal detective once you have enough history (0.64 vs 0.65 at 50 receipts) in this version of the impostor game. The same happened for brand-new accounts. One warning: in the online shop the crowd detective was barely better than guessing, so starting from the crowd made the blend *worse* for the first few receipts. A blend only helps if the crowd knows something useful.

There is a catch we found when checking our own work. After the crossover, the blend (with the setting we chose in advance) was often a bit *worse* than the "you only" detective: when we swapped only receipts customers made themselves, when we made amounts five times bigger, and when we scored the detectives a different way. Part of the reason is that the blend measures you on the crowd's scale instead of your own.

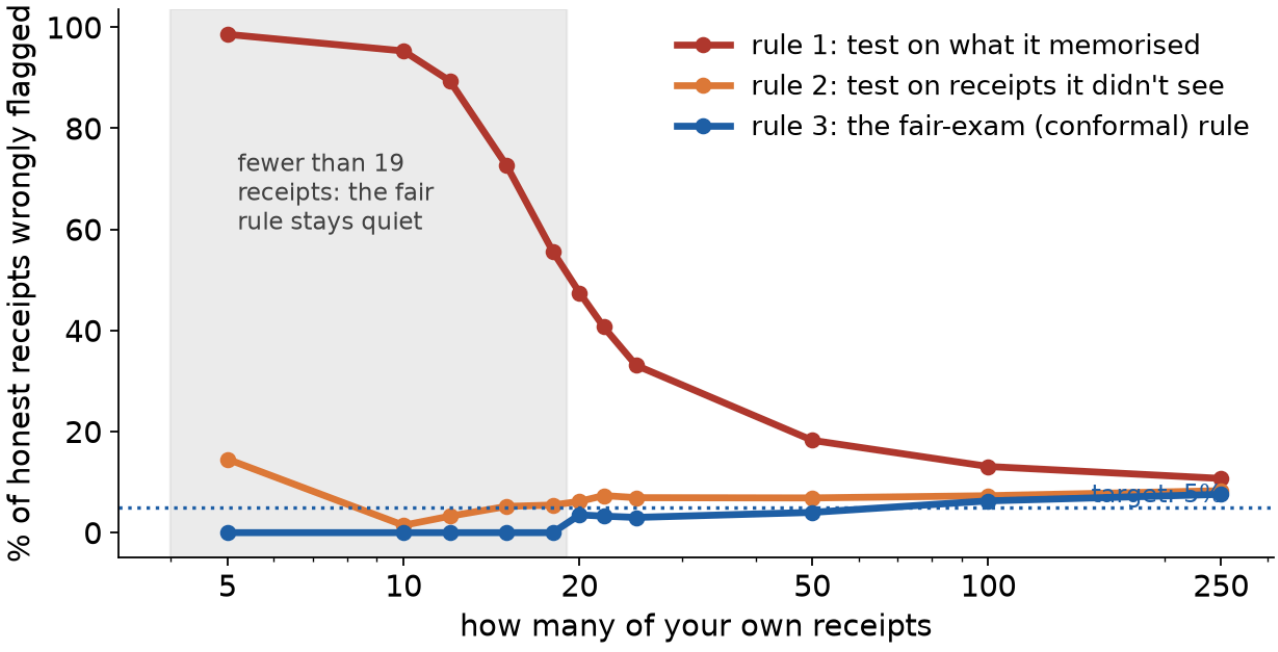
> **Lesson:** don't put a brand-new customer straight onto "personal-only" monitoring. Start with the crowd (or the blend), and switch to the personal detective once the person passes the crossover. First check, on other customers, that the crowd detective is actually useful.

10. Result 3: a good detective still needs a fair alarm bell



Ranking is not enough. A real system must decide: *ring the alarm or not?* We want the alarm to ring for about 1 honest transaction in 20 (5%). We tried three ways to set the alarm level from the person's own past:

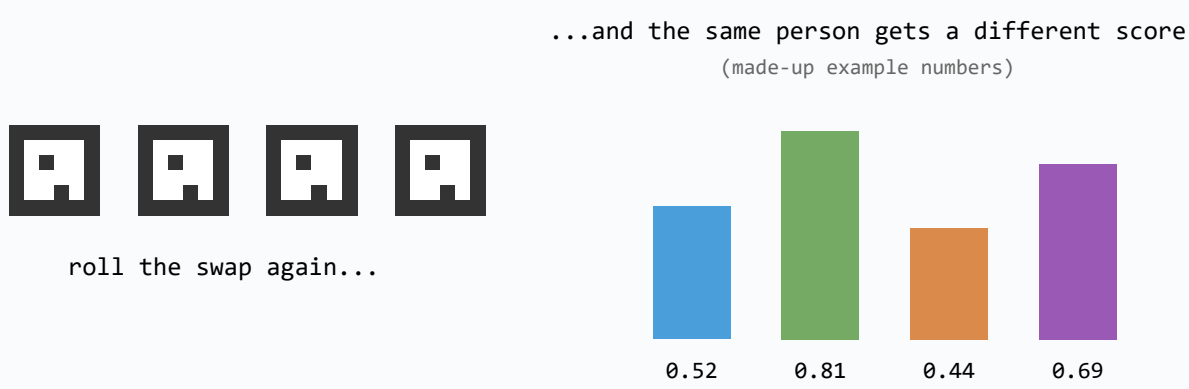
1. **Test it on what it memorised.** The detective has already seen these receipts, so they all look very normal, the bar is set too low, and the alarm rings for almost everything (95% of honest receipts at 10 receipts).
2. **Test it on receipts it didn't learn from** (leave one out). Much better, but with few receipts it is either too quiet or too noisy.
3. **The fair-exam rule (called "conformal").** It uses a simple counting fact: with fewer than **19** past receipts you cannot promise "at most 1 in 20". So the rule stays quiet until it can keep that promise.



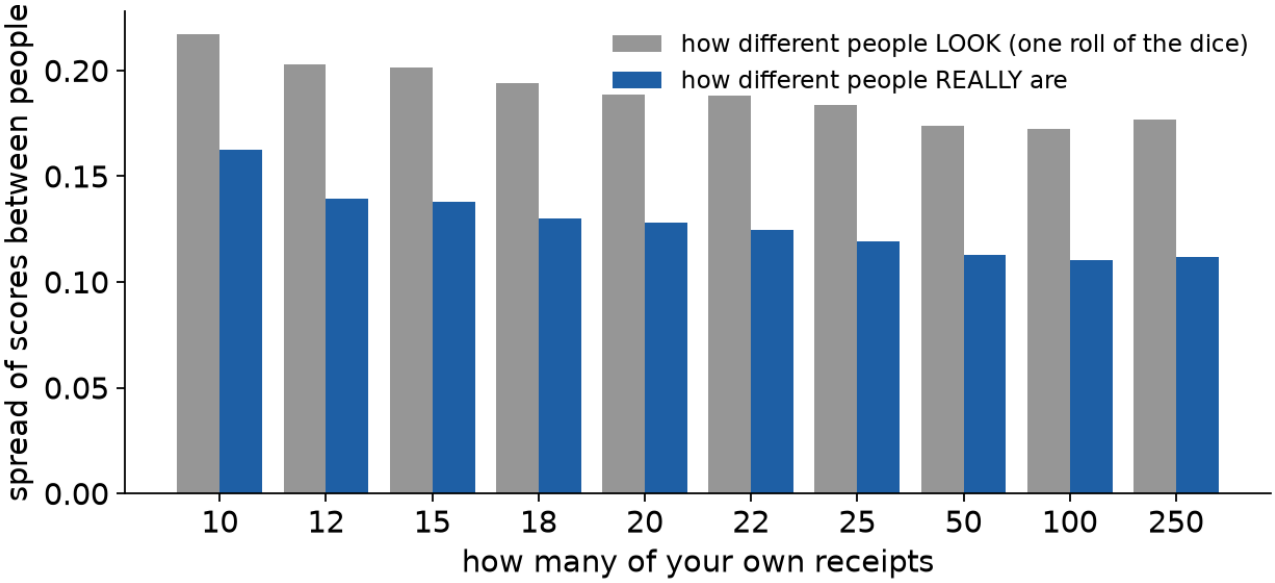
With the fair rule and 20-50 receipts the alarm rang for 3.0%-4.0% of honest receipts (target 5%) and caught 32%-47% of impostors. With 100-250 receipts it rang a bit *too* often (7.6% at 250). A likely reason: very long histories reach years back, and old habits are not quite today's habits.

The promise has small print. The fair rule's promise only holds if your past receipts look like your present ones. In the online shop, one shopping trip makes many receipt lines at once, so 25 lines are really only about 2 shopping trips. There the rule flagged 37% of honest lines with 25 lines of history instead of 5%. On the card data it went the other way and almost never rang with 25 receipts.

11. Result 4: the dice problem (and real differences between people)



Each person's test pile has only 6 swapped receipts. Which 6 get swapped is luck, like rolling dice. Roll again and the same person's score changes. We rolled the impostor game **20 times** for every person, and measured how much of the difference between people is just dice:



With one roll, scores between people spread by about 0.18. After removing the dice noise, the **true** spread is about 0.12. So roughly 60% of what *looks* like differences between people in a single roll is luck.

But real differences remain: people genuinely differ, and that spread barely shrinks between 25 and 250 receipts. For 25% of accounts, adding their oldest receipts made the detective noticeably *worse* again, so they never settle. Among the rest, half reach their own 90%-of-best point by about **50** receipts and a quarter need 100 or more. So the average answer (20) can be too optimistic for individuals (part of that gap may come from how noisy one person's curve is).

Can we predict who needs more history?

In the **real bank data**, no: none of the 8 "messiness" measures predicted who needs more, even though our measurement was reliable enough to spot a modest link (it could detect a true correlation of about 0.21).

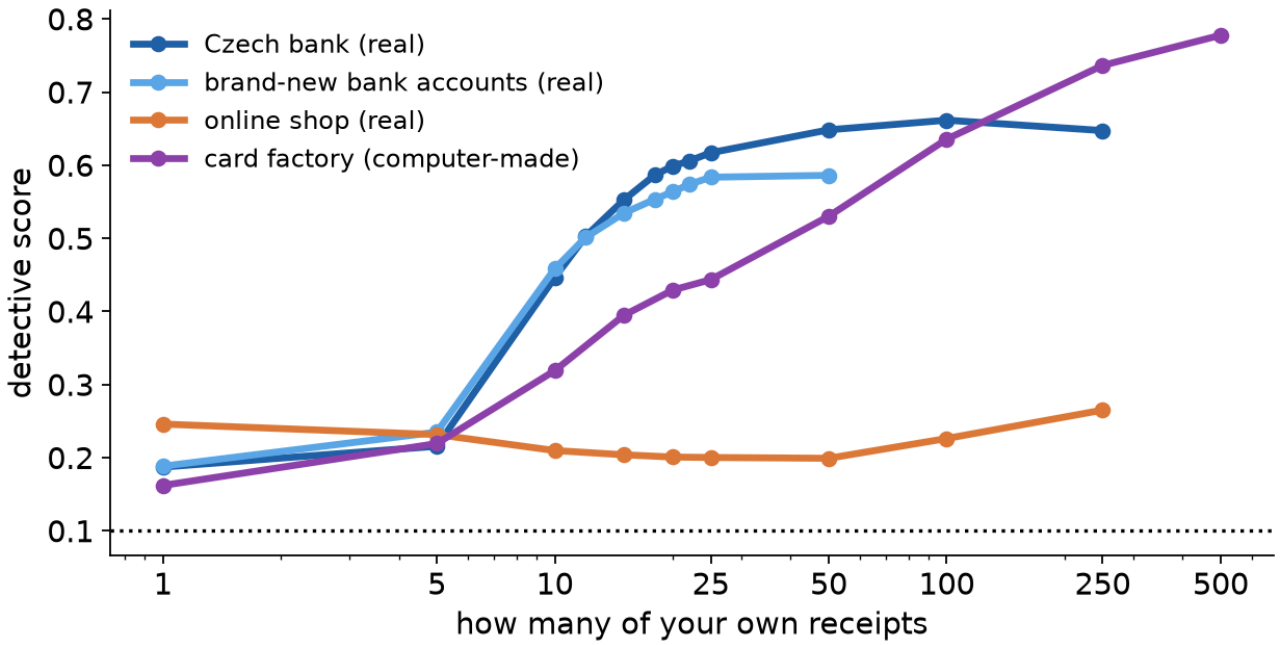
In the **sandbox** we set how messy each *type* of person is, and messy types really do need far more receipts. But inside one type everyone got exactly the same messiness settings, so the sandbox cannot tell us whether messiness would predict differences between *individual* people. That question stays open.

Do old receipts hurt because habits change ("drift")?

Habits do drift: older receipts look measurably different from recent ones. But how much a person drifted did not predict whether adding their old receipts helped or hurt (correlation -0.09, and still small after correcting for the dice noise: -0.09). So drift is real but is not the explanation we can confirm.

We also kept the number of receipts fixed at 50 and changed *which* 50 the detective saw. The 50 most recent were best. Fifty older ones, or fifty spread thinly over a longer time, gave lower scores. Recent history beats old history.

12. Result 5: does it hold in other worlds?



The Mahalanobis detective in each world.

World	Best detective there	Its best score	Receipts to reach 90% of its best
Czech bank, long histories	Mahalanobis	0.66	20 to 22
Czech bank, brand-new accounts	Mahalanobis	0.59	15 (still improving at the end)
Online shop	amount only	0.37	1
Card factory (made-up)	Mahalanobis	0.78	250 (still improving at the end)

- **Brand-new bank accounts** look like the long-history accounts: about 15 receipts, and the crowd wins below about 10.
- In the **online shop** the best clue is simply how much someone spends per item, and **one** receipt line already shows it. The fancier detectives hardly learn at all, because receipt lines come in shopping-trip bunches.
- In the **card factory**, which has shop names and places, the detectives keep improving even at 500 receipts. There is no "enough" within the data we had.

So the honest answer is not a single magic number. It depends on the kind of spending and on the detective, and a bank would need to measure it on its own data, which this

project's method makes possible.

13. Result 6: the sandbox

In the sandbox we can make people tidy or messy on purpose. Tidy people reach 90% of their best within about 50 receipts; travellers and very varied people need hundreds. They all end up at a similar best score: messiness changes *how fast* the detective learns, not how good it finally gets.

The sandbox also fixed a mistake in an earlier version of this study. A blend computed against the crowd's list of shops scored only 0.45 at 500 receipts, but a blend that remembers *your* shops scored 0.92, as good as the personal detective (0.92). So blending works, as long as the blend is allowed to remember what is personal about you.

14. What we can and cannot conclude

We can say

- On real bank data, a personal detector reaches 90% of its best after 10 to 22 transactions and improves only a little after 50.
- On bank data, below about 10 transactions a crowd model is better; after that, the personal detector is at least as good.
- A fair alarm needs its own rule and at least 19 transactions; ranking well is not enough.
- Individuals really differ, and the average hides it.
- The number changes with the kind of data and the detective, so every system should measure its own.

We cannot say

- That any of this catches fraud. Our "unusual" receipts are swapped-in honest receipts from other people.
- That it holds for every bank today. The main real data are from one Czech bank in the 1990s.
- That it works against a clever criminal who copies your habits. We did not test that.
- That the bias is fair across ages, genders or backgrounds. The data had no such information about individuals.

15. Being careful (ethics)

- **Different is not suspicious.** Life changes (a new job, a move, an illness) look "unusual". A system should never treat a person as guilty because they changed.
- **False alarms are not free.** Even when the fair alarm keeps its promise (about 3%-4% of honest payments flagged with 20-50 receipts), a big bank would still wrongly flag a great many honest payments every day. Someone making 40 payments a month would get one or two false alarms a month.

- **New customers need protection.** On the bank data, personal-only monitoring serves people with short histories worst, which is why starting from the crowd and using a fair alarm matter.
- **Privacy.** On the bank data, recent history was enough and old history did not help, so keeping less data about people did not cost accuracy there.
- We used only public, anonymised data and never tried to identify anyone.

16. Glossary

Anomaly	Something unusual compared with what came before. Not the same as fraud.
Personal history size (*n*)	How many of a person's own past transactions the detective learned from.
Impostor substitution	Swapping a real receipt for another person's real receipt, so we know which ones are "not you".
PR-AUC	A 0-to-1 score of how well a detective ranks the swapped receipts near the top. Guessing gives 0.10 here.
Sufficiency point	The history size where the detective reaches 90% of its best score.
Crowd (population) model	A detective trained on other people.
Hybrid / blend	A detective that mixes the crowd and the person, trusting the person more as history grows.
False-positive rate	The share of honest transactions wrongly flagged.
Conformal rule	A way to set the alarm level with a promise: "at most 1 in 20 honest transactions flagged", valid when the past looks like the present.
Reliability / ICC	How much of the difference between people is real rather than dice-roll luck.
Seed	A starting number for the random choices, so anyone can repeat exactly the same experiment.

17. Check our work

The project lives at `github.com/briyandyju09/personal-history-anomaly-detection`, which holds the code and results of the first version. The updated code and result tables behind this explainer are available from the author on request. A built-in test re-runs an older version of the pipeline and checks that it gives the same numbers to 15 decimal places before any new result is trusted.

This explainer was generated from the same result file as the research paper. The paper has the full details, statistics and references. Text and illustrations were produced with the help of an AI assistant (Claude, by Anthropic).